

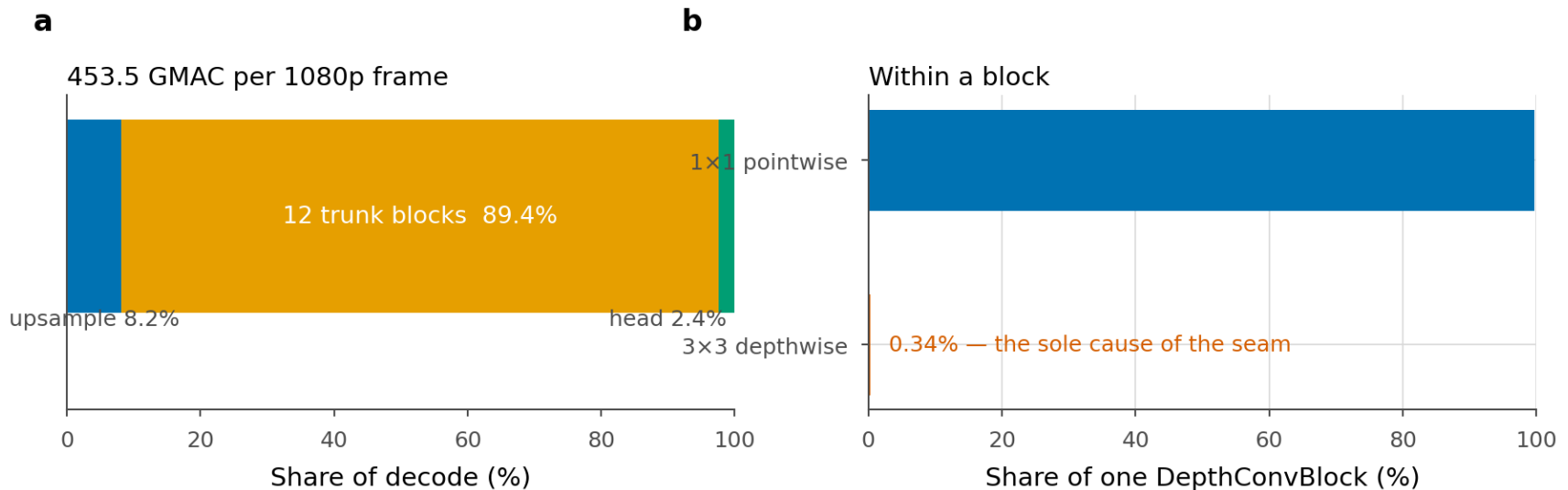


FLEX-UF: Content-Adaptive Early Exit for DCVC-UF

Decode compute spent where the content needs it
Chair of Media Technology · TUM

Uniform compute, non-uniform content

- **DCVC-UF intra decoder · 453.5 GMAC / 1080p frame**
 - upsample 8.2% · 12 DepthConvBlocks 89.4% · head 2.4%
- **Target · 30–40% saved at ≤ 0.1 dB**
- **Constraint · encoder, hyperprior, entropy model frozen**
 - coded payload bit-identical → the decoder is the only variable

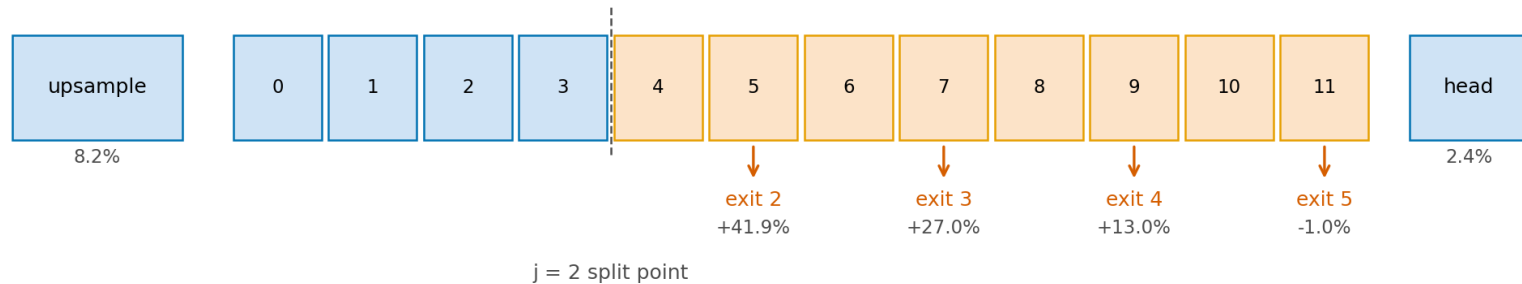


A ladder of exits over the trunk

- **K exits over 12 blocks, split depth j**
 - groups $0 \dots j-1$ full-frame · groups $j \dots K-1$ per tile
 - shipped: $K = 6$, $j = 2$, 256 px tiles
- **Per-tile cost C_k (units of one stock decode)**
 - 0.581 0.581 0.581 0.730 0.870 1.010
 - exits 0,1 end inside the stem $\rightarrow C_0 = C_1 = C_2 \cdot 6$ rungs, 4 prices

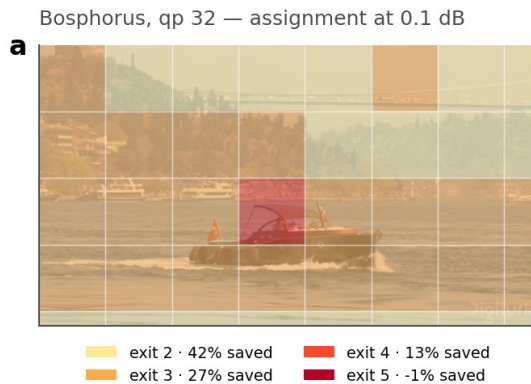
full-frame — no seams

per tile — the skippable part, 89.4% of the decode



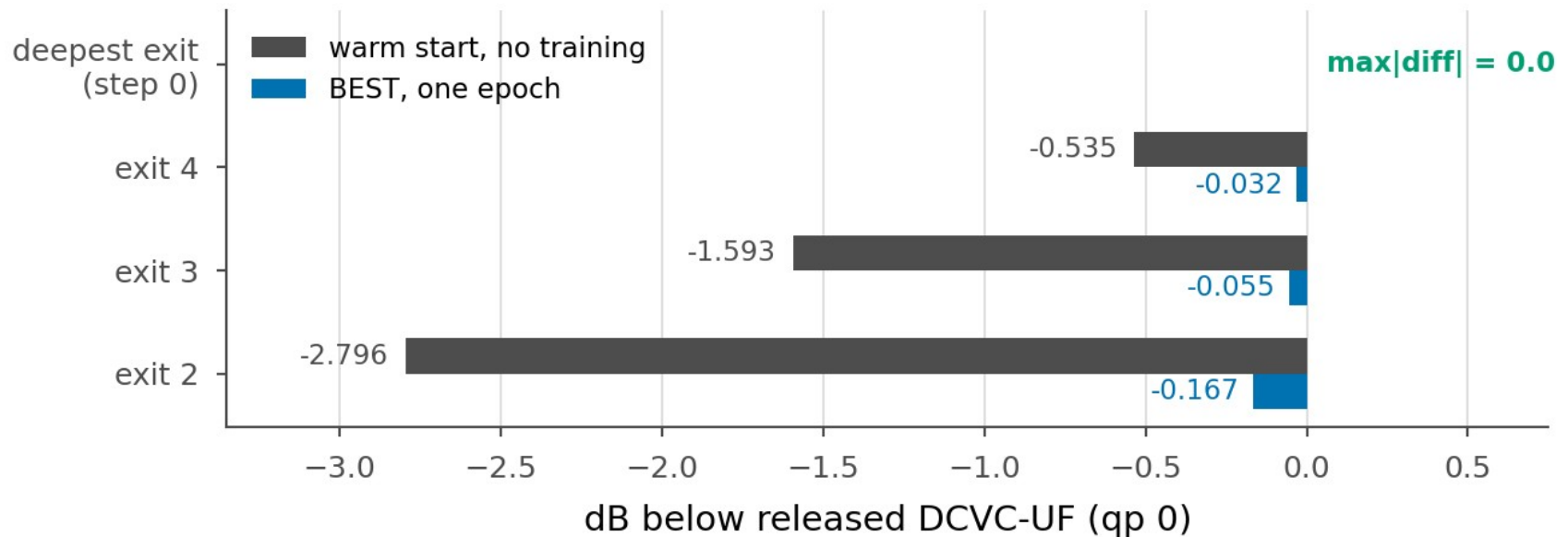
The assignment, on a frame

- **Bosphorus · qp 32 · λ^* bisected to 0.1 dB · 40 tiles**
- **boat → exit 4 water, sky → exit 2 shoreline → exit 3**
 - 33.4% saved at 0.098 dB
 - (c) dB vs each tile's OWN reference, not the frame mean



The reference IS the released decoder

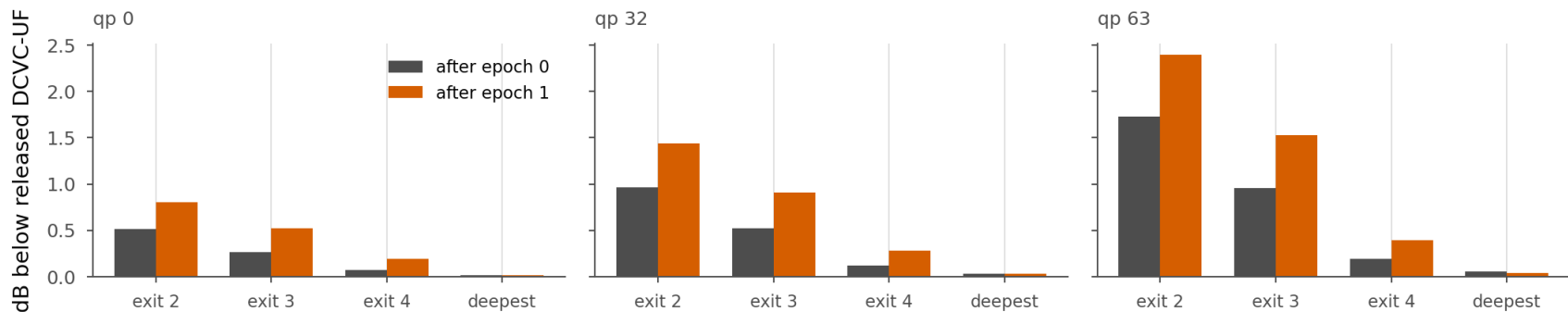
- Released weights re-expressed in ladder form · bijection over 143 tensors
- Adapters and seam repair zero-init $\Rightarrow$ identity at step 0
- $\max |\hat{x}_{\text{release}} - \hat{x}_{\text{ours}}| = 0.0$ at qp 0 / 32 / 63
 - through Microsoft's own DMCI class and forward_one_frame
- So training must (a) lift shallow exits, (b) not damage the deepest



Recipe, checked against theirs

- **verify_recipe.py** diffs each item against `~/DCVC/train_image.py`
- **Deepest-exit drift vs release** BEST -0.003 / -0.016 / -0.033 · VERBATIM (no anchor) -0.119 / -0.132 / -0.174 at qp 0/32/63
 - Saving at 0.1 dB, same architecture: anchor 0: unreachable at every rate · anchor 1: 3/3/2/2/2% · anchor 10: 42/42/42/42/42% (CONTROL → BEST moves four flags, so it is the bundle)
 - CONTROL, 2nd epoch: saving at qp0 13.3% → 3.8% · damage grows the shallower the exit · VERBATIM improved over the same boundary → cause unknown
- **Additions, listed:** `--epoch_offset` · `--anchor_weight` · `--new_lr_scale` · `--freeze_encoder`

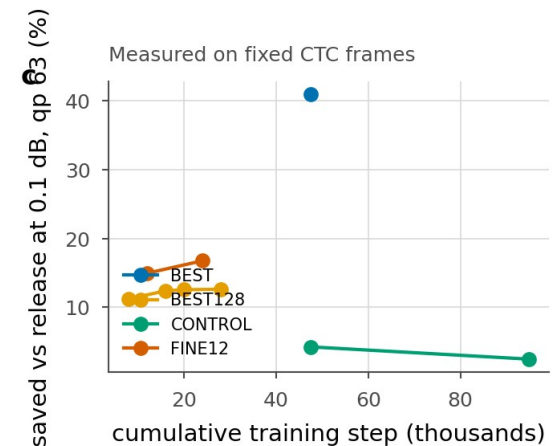
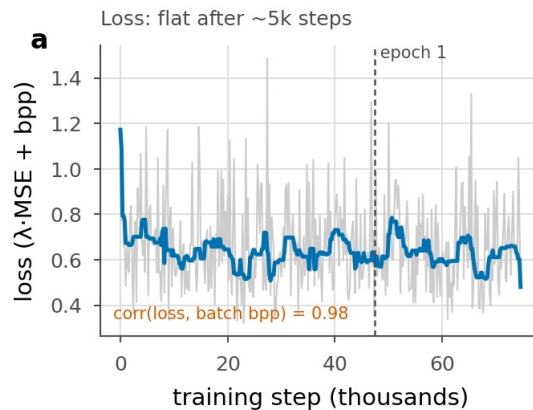
CONTROL: a second epoch degrades every exit except the deepest. VERBATIM, with fewer additions, improved — cause open



The loss is not the convergence signal

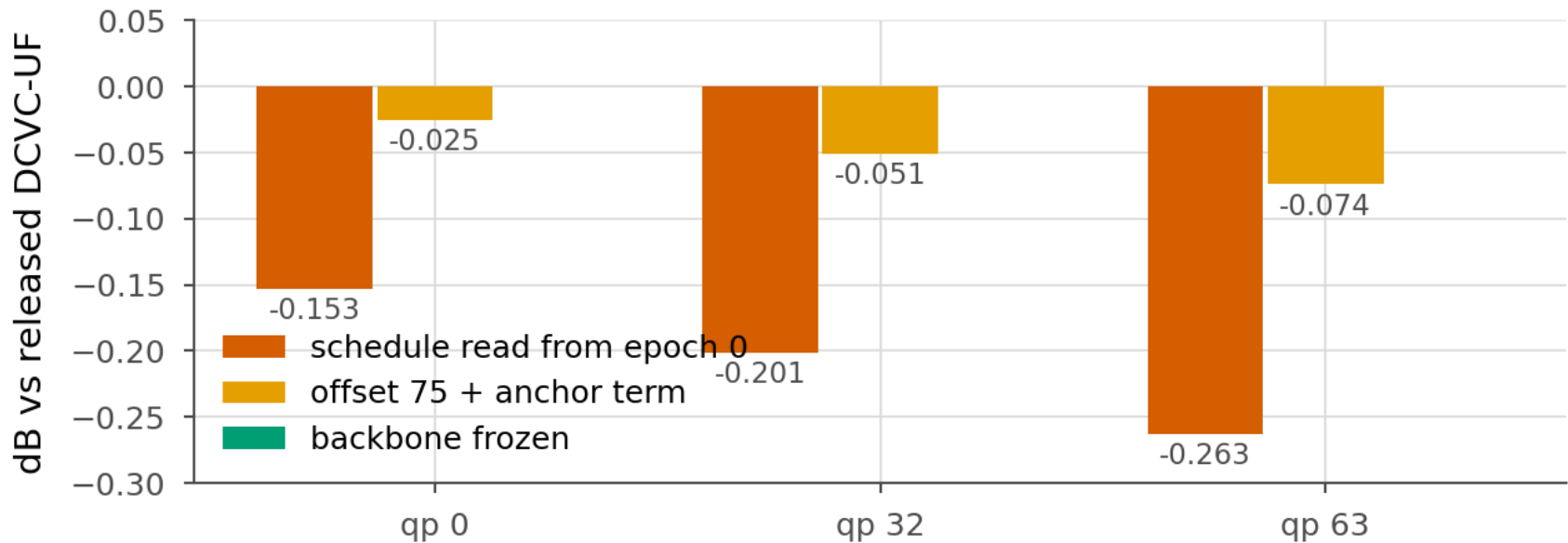
- $L = \lambda \cdot \text{MSE} + \text{bpp}$ · $\text{corr}(L, \text{batch bpp}) = 0.97$
 - flat from step ~5,000; a 256 px crop's bitrate swings 2× per batch
- (b) exits stay ordered, ~1 dB apart → ladder healthy
- (c) saving on FIXED CTC frames, still rising at 20–24k steps
 - BEST128 11.2 → 12.6% FINE12 15.0 → 16.8% (qp 63)
- 1 epoch = 47,451 steps · every number here is one checkpoint

BEST: one epoch is 47,451 steps



Schedule position, and a $\times 4$ error

- **Reading the schedule from epoch 0 applies from-scratch lr to a converged model**
 - deepest exit fell 0.153 / 0.201 / 0.263 dB in one epoch
 - $\rightarrow$ 0.025 / 0.051 / 0.074 with --epoch_offset 75 + --anchor_weight
- **Freezing the trunk gives 0.000 drift — and collapses the ceiling 27.3% $\rightarrow$ 11.5%**

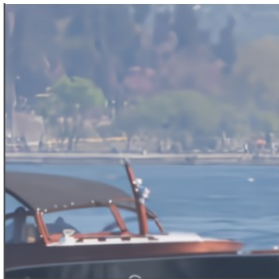


One frame, four exits

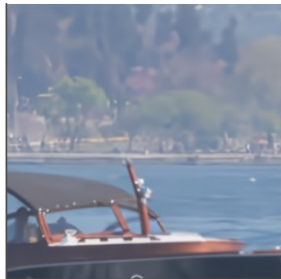
- **Bosphorus 1080p · qp 32 · identical bitstream at every exit**
- **Error ×25 · damage is structured: detail and tile borders**

Bosphorus 1080p, qp 32 — identical bitstream, different exits

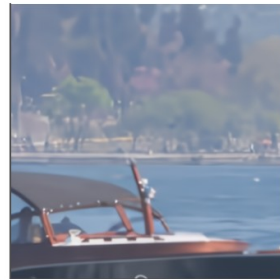
Released DCVC-UF



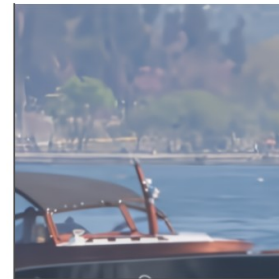
Exit 2
42% saved, +0.232 dB



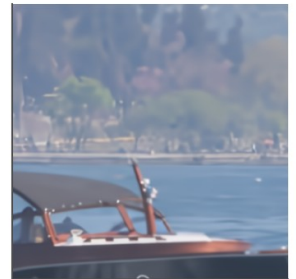
Exit 3
28% saved, +0.077 dB



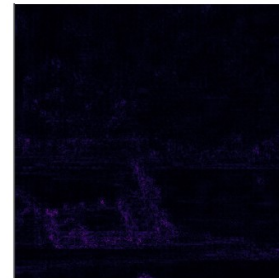
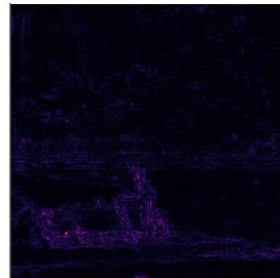
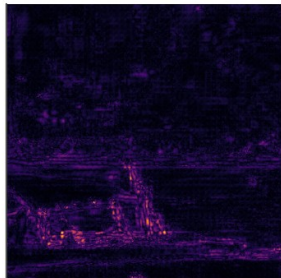
Exit 4
14% saved, +0.045 dB



Exit 5
0% saved, +0.013 dB



Error maps
(amplified ×25)



Seam 1/5 · where it comes from

- **3×3 depthwise at (i, j):**
 - $y[i,j] = \sum_{\{u,v \in \{-1,0,1\}\}} w[u,v] \cdot f[i+u, j+v]$
- **Per tile, $f[i+u, j+v]$ outside the tile does not exist → padding invents it**
 - only the 3×3 depthwise has spatial extent; everything else is 1×1
 - ⇒ exit adapters are 1×1 on purpose: no receptive field, no seam
- **Contamination grows one ring per layer. b per-tile layers, tile side F:**
 - corrupted fraction = $1 - ((F - 2b) / F)^2 \approx 4b / F$
 - $F = 32, b = 8 \rightarrow 1 - (16/32)^2 = 75\%$ of the tile
- **Not a line of bad pixels — the error propagates inward**
- **Scales with tile PERIMETER, and with split depth j**

Seam 2/5 · padding is an estimator

- **Border column $x[0]$, unseen neighbour $x[-1]$:**
 - ☐ zeros $\hat{x}[-1] = 0$
 - ☐ replicate $\hat{x}[-1] = x[0]$ zeroth-order hold
 - ☐ linear $\hat{x}[-1] = 2 \cdot x[0] - x[1]$ first-order
 - ☐ arls $\hat{x}[-1] = a \cdot x[0]$, a fitted per channel AR(1)
- **Pure seam penalty · CTC · 256 px · full depth · dB above the release**

	qp 0	qp 32	qp 63
zeros	0.1005	0.2162	0.5477
replicate	0.0616	0.0854	0.1070
linear	0.1793	0.2315	0.2638
arls	0.0518	0.0707	0.0879
- **linear > replicate: extrapolating a gradient amplifies boundary noise**
- **arls wins on dB, REJECTED on cost · +10.7% wall-clock for 0.019 dB**

Seam 3/5 · the perimeter law

- **Prediction $4b/F \Rightarrow$ doubling F halves the penalty**
 - ☐ zeros qp 63: 128 px 1.1670 $\rightarrow$ 256 px 0.5477 $\times 0.47$
 - ☐ replicate qp 63: 128 px 0.2125 $\rightarrow$ 256 px 0.1070 $\times 0.50$
- **Free in compute — a tiled decode's MAC count is independent of F**
- **Paid in routing granularity · 40 tiles/frame at 256 px vs 160 at 128**
- **Halo, same law: tile F with halo h computes $((F+2h)/F)^2$ pixels**
 - ☐ $F = 16, h = 4 \rightarrow 2.25\times$ $F = 32, h = 4 \rightarrow 1.56\times$
 - ☐ $\Rightarrow$ halo on the HEAD only (2.40% of MACs); on the trunk it exceeds the saving

Seam 4/5 · repair told where to look

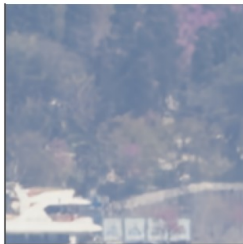
- **A translation-invariant 3×3 must INFER the boundary from content**
 - and corrects the ~25% clean interior, where any correction is damage
- **But the lattice is known — unpatchify lays tiles on a fixed P×P grid**
- **Repair(f) = f + G[i mod P, j mod P] · PW(WSiLU(DW3×3(f)))**
 - G : P × P = 256 scalars, shared over all 384 channels
 - 0.0007% of decoder parameters, no MAC beyond a broadcast multiply
 - $G_0 = \exp(-d / \tau)$, d = distance to nearest tile edge
 - $\Rightarrow$ the gate starts on the seam; training refines, not discovers
 - PW zero-init $\Rightarrow$ module starts as identity $\Rightarrow$ warm start stays bit-exact
- **Cost 0.95% of the decode — charged inside every saving, incl. the deepest exit ($C_{\{K-1\}} = 1.0095$, not 1.0)**

Seam 5/5 · at pixel scale, and what repair is

- 192 px across a tile boundary · qp 63 · full depth · error ×40
- Repair ON vs OFF, 8 sequences: full depth 0.0672 → 0.0750 (worse) · routed 0.1259 → 0.1239
- 0.0020 dB/point of decode · arls was rejected at 0.0018
 - ☐ trained jointly ⇒ an inference ablation is not the clean test

Bosphorus, qp 63 — every tile at FULL depth, so the only difference is the tile border (dashed line)

Full-frame decode
(no tiling at all)



zeros padding
+0.113 dB, whole frame



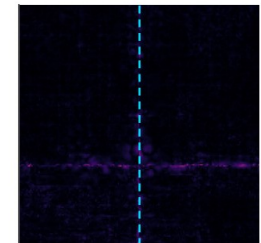
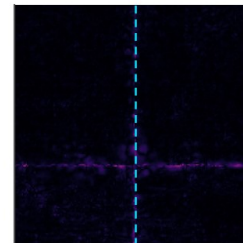
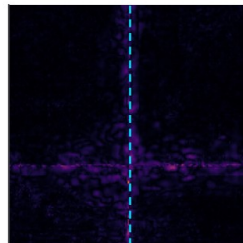
replicate padding
+0.072 dB, whole frame



replicate + grid repair
+0.080 dB, whole frame

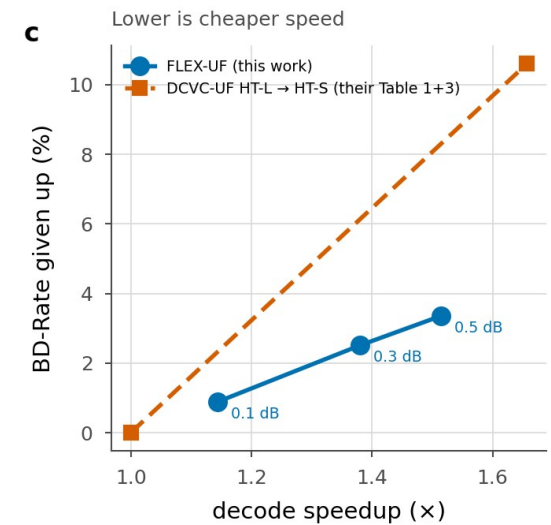
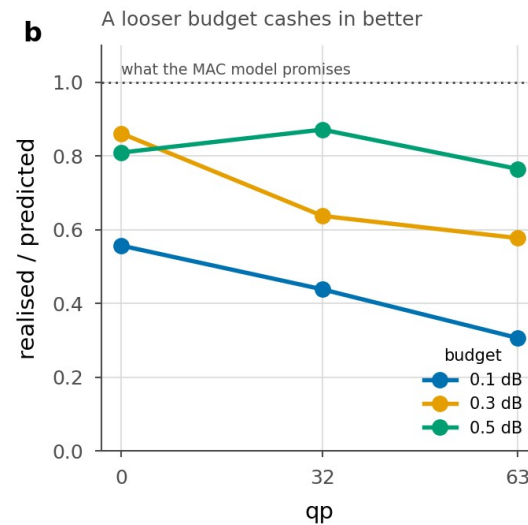
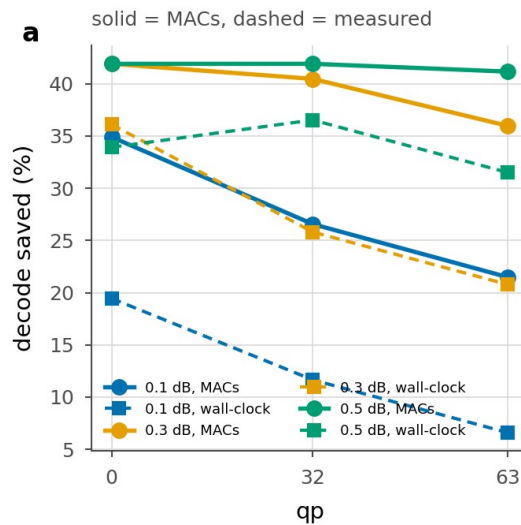


|error| vs full-frame
amplified x40



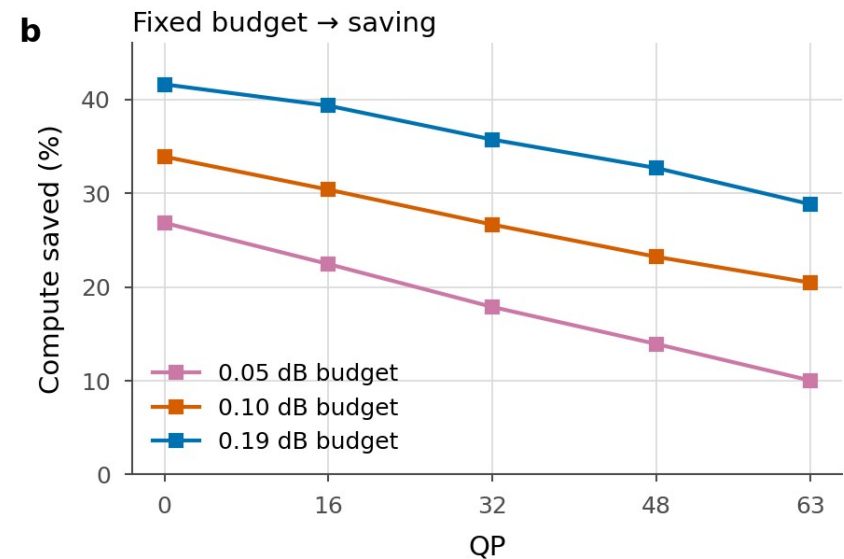
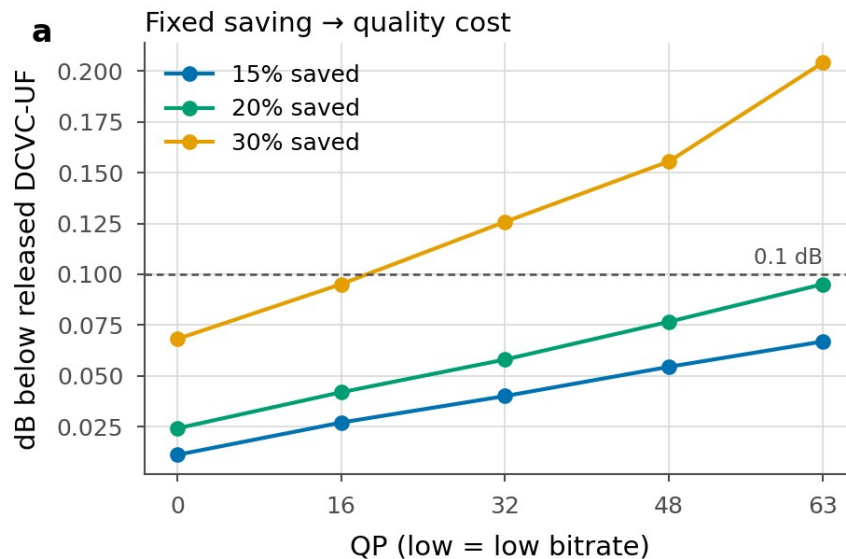
MACs are not milliseconds

- **1920×1088 · interleaved · median of 40**
 - 0.1 dB: 28% of MACs -> 13% of wall-clock (1.14x)
 - 0.3 dB: 39% of MACs -> 28% of wall-clock (1.38x)
 - 0.5 dB: 42% of MACs -> 34% of wall-clock (1.51x)
- **Gap = tiling overhead 3.7–6.5% + per-group bookkeeping**
 - mask + gather + scatter + one device ↔ host sync per group
 - 36.4 ms vs 77.5 ms of convolution = 32% of the loop
- **Recoverable · sort by exit depth once → 23–37% back, bit-identical**



Against the released decoder

- At 0.1 dB · 34.0% · 26.8% · 20.8% (qp 0 / 32 / 63)
☐ ÷ RELEASE (1.0), not our deepest exit (1.0095) — earlier decks read 0.6–0.75 pts higher
- The 30% target is cleared by qp 0/16 at 0.1 dB, and by qp 32/48/63 at 0.3 dB
- Ladder ceiling 41.9% at every rate, set by the split depth $j=2$, not by training
- BD-saving 30.8% over dB in [0.064, 0.196] (qp 0: 37% · 16: 34% · 32: 31% · 48: 27% · 63: 24%); BD-quality 0.062 dB over saving in [10%, 30%]



The allocation problem

N tiles, K exits. $D[t,k]$ = distortion of tile t at exit k. C_k = its cost.

$$\min_{k_1 \dots k_N} \frac{1}{N} \sum_{t=1}^N D[t, k_t] \quad \text{s. t.} \quad \frac{1}{N} \sum_{t=1}^N C_{k_t} \leq B$$

Lagrangian relaxation decouples it — every tile solves independently:

$$k_t^*(\lambda) = \arg \min_k \{ D[t, k] + \lambda C_k \}$$

sweeping λ traces the lower convex hull of the achievable (cost, distortion) set

λ is found by BISECTION on the realised dB, so a 0.1 dB budget is exactly 0.1

Both configurations solve this. They differ only in where $D[t,k]$ comes from.

Two ways to obtain the decision

A · the encoder holds the source, so $D[t,k]$ is measured, and k^* is exact

$$D[t, k] = \| \hat{x}_k[t] - x[t] \|_2^2 \quad \Rightarrow \quad k_t = k_t^*(\lambda)$$

the answer is then transmitted: $H(k) \cdot N + 8K \text{ bits} \approx 94 \text{ bit/frame}$

B · the decoder never sees x , so $D[t,k]$ does not exist. It predicts instead.

$$z_t = \text{MLP}\left(\left[\mu_t, \sigma_t, \rho_t, \frac{qp}{63}\right]\right)$$

μ, σ : mean and std over tile t of a learned $384 \rightarrow 48$ view of the stem

ρ : a $512 \rightarrow 32$ view of the latent $\hat{y}$ concatenated with the entropy scales $\hat{\sigma}$

$$\hat{k}_t = \arg \max_k \{ \log \text{softmax}(z_t)_k - \beta C_k \}$$

β plays λ 's role, but trades log-likelihood against cost, not distortion against cost

Training the router, and why it collapsed

Target is the oracle's choice, weighted by what the decision is worth:

$$\mathcal{L} = \text{CE}_w(z_t, k_t^*) + \alpha \mathbb{E}_t[L(t, \hat{k}_t) - L(t, k_t^*)], \quad L(t, k) = D[t, k] + \lambda C_k$$

agreement on ties is worth nothing; the regret term says so

Collapse control without an auxiliary loss (arXiv:2408.15664):

$$z_t \leftarrow z_t + b, \quad b \leftarrow b + \eta (\pi^* - \hat{\pi})$$

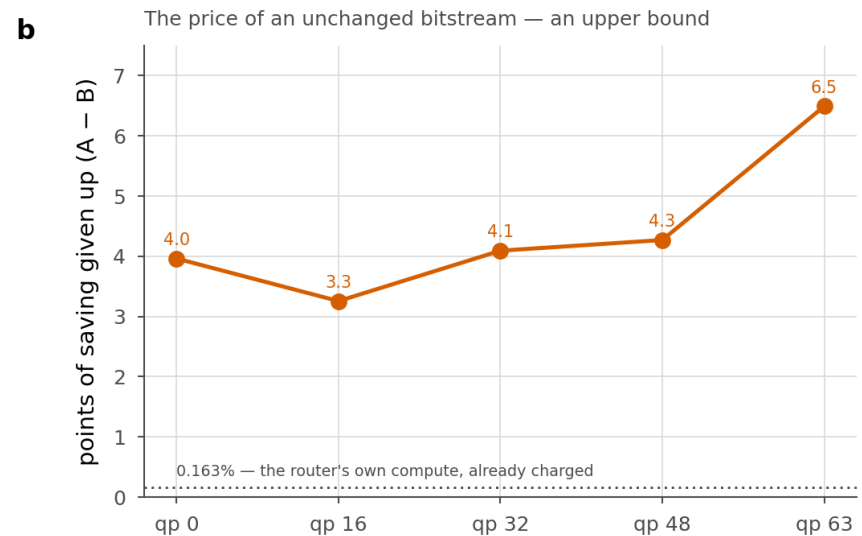
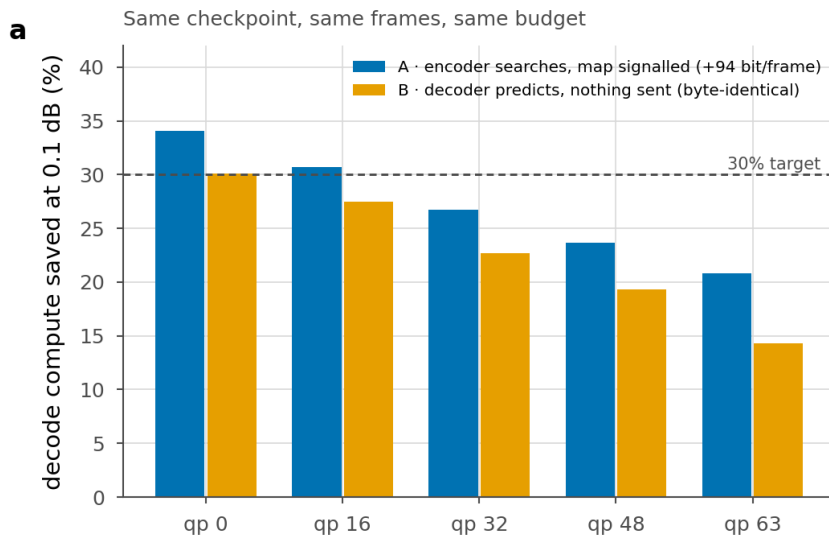
b is a buffer, nudged by usage — no interference gradient enters $\mathcal{L}$

balanced toward the ORACLE's mix π^* , not uniform: exits are not interchangeable experts

Trained JOINTLY with the decoder → collapsed to a constant, agreement 0.000 at qp63. Against the FROZEN decoder → 0.848.

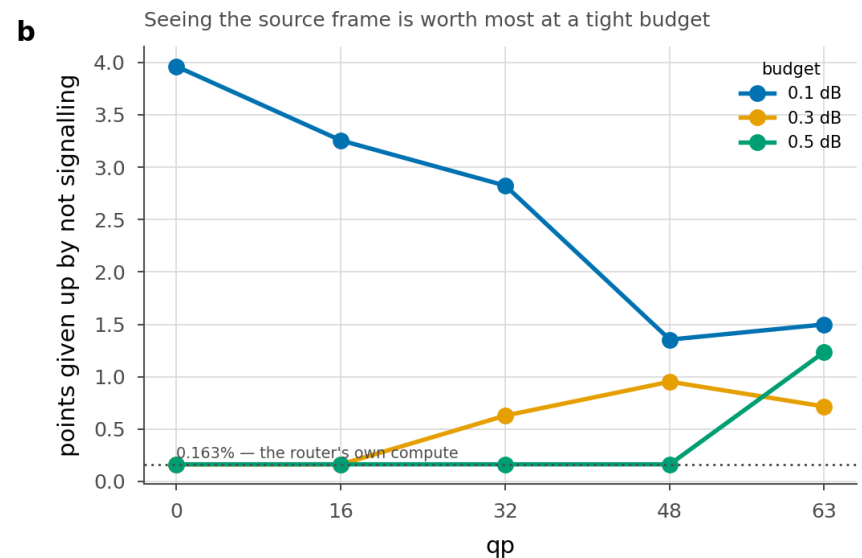
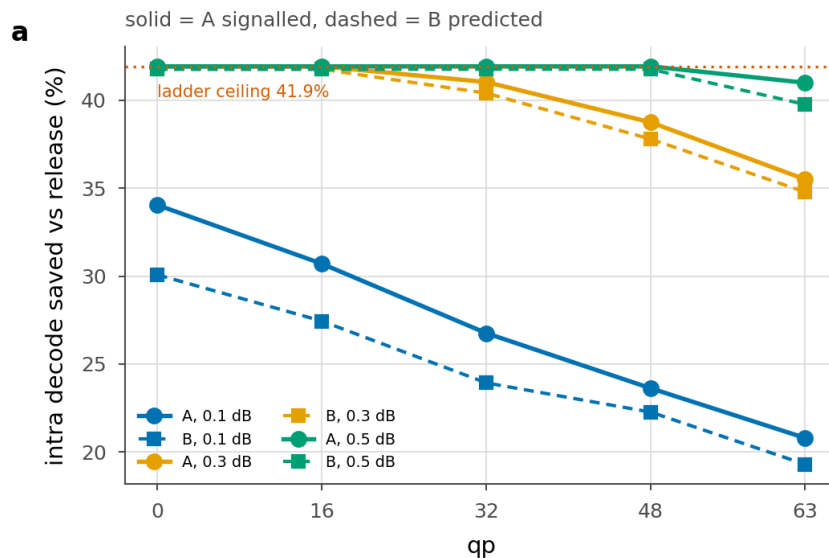
Who decides

- **A** $\text{argmin}_k (\text{MSE}[t,k] + \lambda \cdot C_k)$ ~94 bit/frame = 0.008–0.020% of rate
- **B** $\text{argmax}_k (\log \text{softmax}(z)_k - \beta \cdot C_k)$ nothing sent, file byte-identical
 - qp0: A 34.0% signalled vs B 30.1% predicted — gap 3.96 pts
 - qp32: A 26.8% signalled vs B 23.9% predicted — gap 2.82 pts
 - qp63: A 20.8% signalled vs B 19.3% predicted — gap 1.50 pts
- **Asymmetric** · **A: +1.21 decodes at the ENCODER** · **B: +0.163% at the decoder**



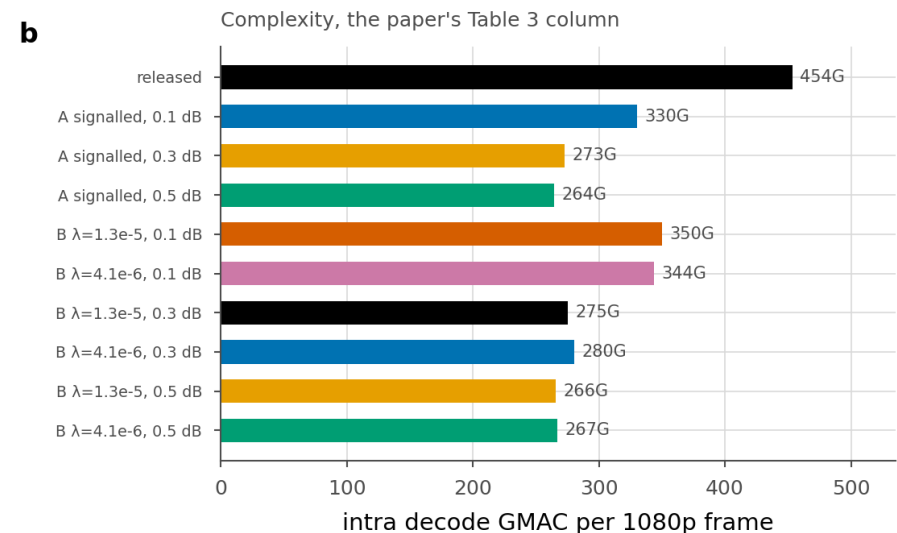
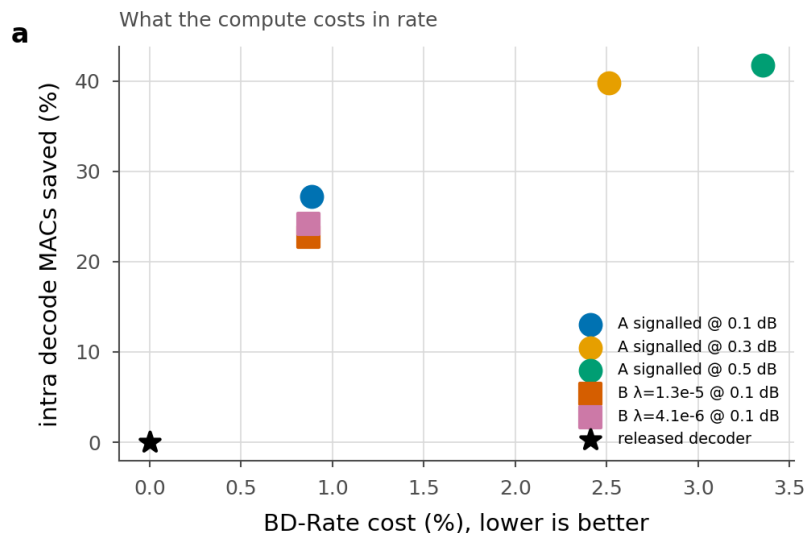
A claim overturned

- **Earlier claim: prediction fails, signalling works**
 - router trained JOINTLY with a moving decoder → collapsed, agreement 0.000 at qp 63
- **Retrained against the FROZEN decoder: 0.848 agreement, budget met at every rate**
 - ⇒ the limit was the training target, not the information
- **Cost of not signalling · 0.1 dB 1.35–3.96 pts · 0.3 dB 0.16–0.95 · 0.5 dB 0.16–1.23**
 - at the ceiling the gap is 0.16 = the router's own 0.163%



The price of speed

- **BD-Rate = cost · MACs saved = benefit · released intra decoder, YUV420**
 - 0.1 dB 0.88% / 27.6% 0.3 dB 2.51% / 39.5% 0.5 dB 3.36% / 41.7%
- **Per unit of REAL speedup: 6.2 · 6.6 · 6.5 BD-Rate points**
 - flat across the range — an exchange rate, not a chosen operating point
- **DCVC-UF HT-L → HT-S: 10.6 points for 1.66× = 16.1 points per unit**
- ⇒ **adaptive depth buys speed 2.4× cheaper than shrinking the model**
 - their pair is whole-video, ours intra-only → a calibration, not a head-to-head

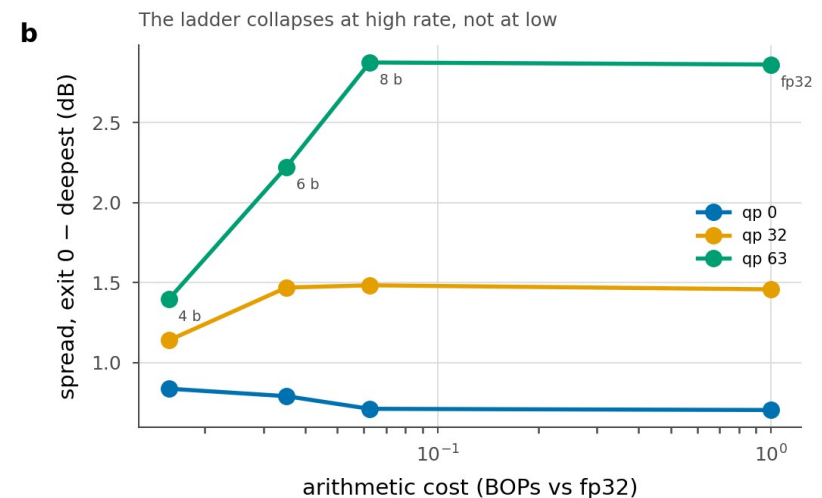
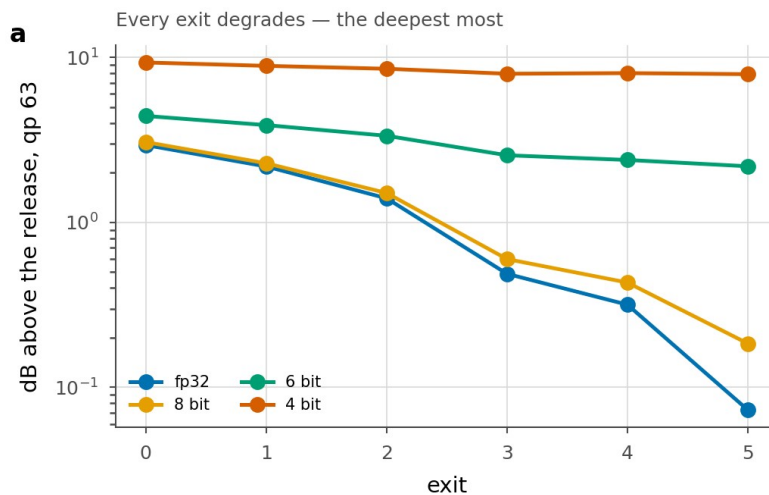


Related work

- **Early exit / dynamic depth · BranchyNet, MSDNet, SkipNet**
 - ☐ decide from the network's own output; ours decides per SPATIAL TILE for a fixed output
- **ClassSR**
 - ☐ routes patches to differently sized SR branches for a cost target
 - ☐ ours: a DEPTH inside one network, for a quality target, bitstream fixed
- **Xie et al. · RDC optimisation in neural image compression · 2305.07678**
 - ☐ adaptive spatial mask decoded from side information — closest to our B
 - ☐ adapts the ENTROPY model's dependencies; we adapt SYNTHESIS depth
- **Spatial competition · 2605.13243**
 - ☐ per-region mode map at $1.8e-4$ bpp (ours $4.5e-5$) — selects CODECS for rate, not depth for compute
- **Loss-free load balancing · 2408.15664 · used directly**
 - ☐ balanced toward the ORACLE's exit mix, not uniform — exits are not interchangeable experts

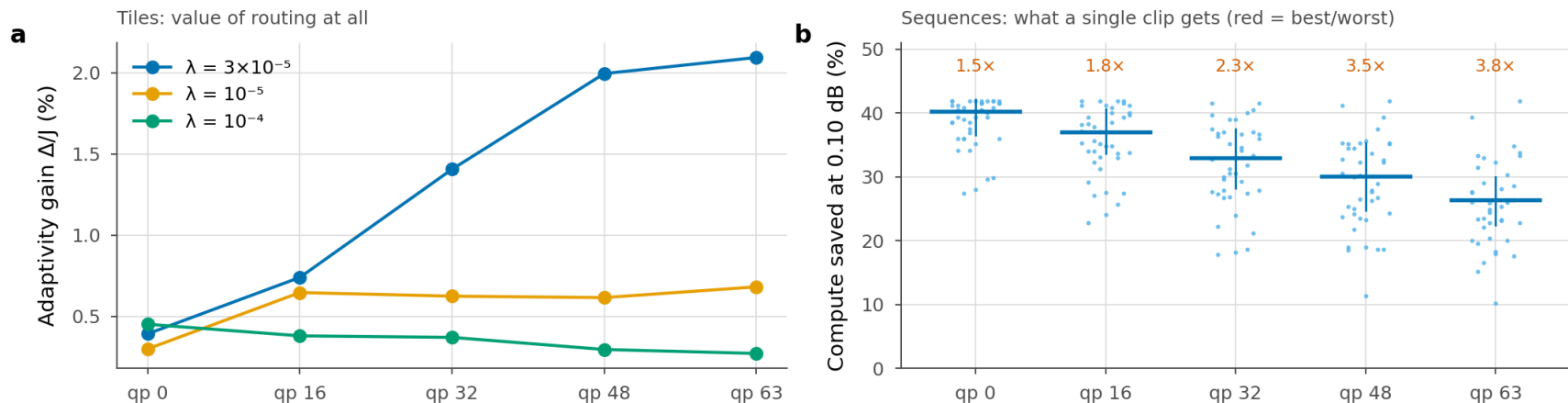
Quantisation · a second, orthogonal lever

- **Weight-only · per-channel · no calibration · decoder only**
- **Registered prediction — shallow exits suffer more — FALSIFIED**
 - qp 63, 8 bit: exit 2 +0.108 dB, deepest +0.111 dB, ratio 0.97
 - the deepest starts at 0.073 dB and has no headroom for a noise floor
- **What matters is the SPREAD — the only thing a router can act on**
 - qp 63 fp32 2.864 → 8 b 2.876 → 6 b 2.224 → 4 b 1.398 · BOPs 1.000 → 0.062 → 0.035 → 0.016
 - qp 0 0.704 → 0.712 → 0.791 → 0.837 — flat to rising, no collapse
- **8 bit composes cleanly at 16× fewer BOPs · the collapse is a HIGH-RATE effect, not a general one**
 - caveat: 8 bit still raises the floor 0.073 → 0.184 dB at qp 63



Why per-tile beats any fixed mix

- **Cost additive over tiles, MSE a mean over tiles $\Rightarrow$ both affine in the assignment**
 - fixed proportions $\rightarrow$ convex hull of the K exit points
 - per-tile assignment $\rightarrow$ Minkowski average of the N tile hulls
- **Gap = min-of-average – average-of-min ≥ 0**
 - zero iff every tile prefers the same exit — this IS the value of adaptivity
- **Measured Δ/J : 0.39% $\rightarrow$ 2.09% with rate**
- **Same mechanism per sequence · best/worst CTC clip 3.8× at qp 63 vs 1.5× at qp 0**



What is still owed

- **A static decoder truncated to the depth we average**
 - ☐ 6.95 of 12 blocks at qp 0 · 8.78 at qp 63
 - ☐ decides whether adaptivity is the contribution
- **All six runs to 4 epochs**
 - ☐ every number here is one checkpoint; the saving is still climbing
- **HEVC B / C / D · 13 of 53 sequences · reported as NOT MEASURED**
- **Sorted-tile execution, end to end**
 - ☐ prototyped and bit-identical; not landed while six runs import the decoder
- **Wall-clock on an idle card · these ratios were taken at 100% utilisation**
- **Open · CONTROL degrades over its second epoch and the cause is unknown**